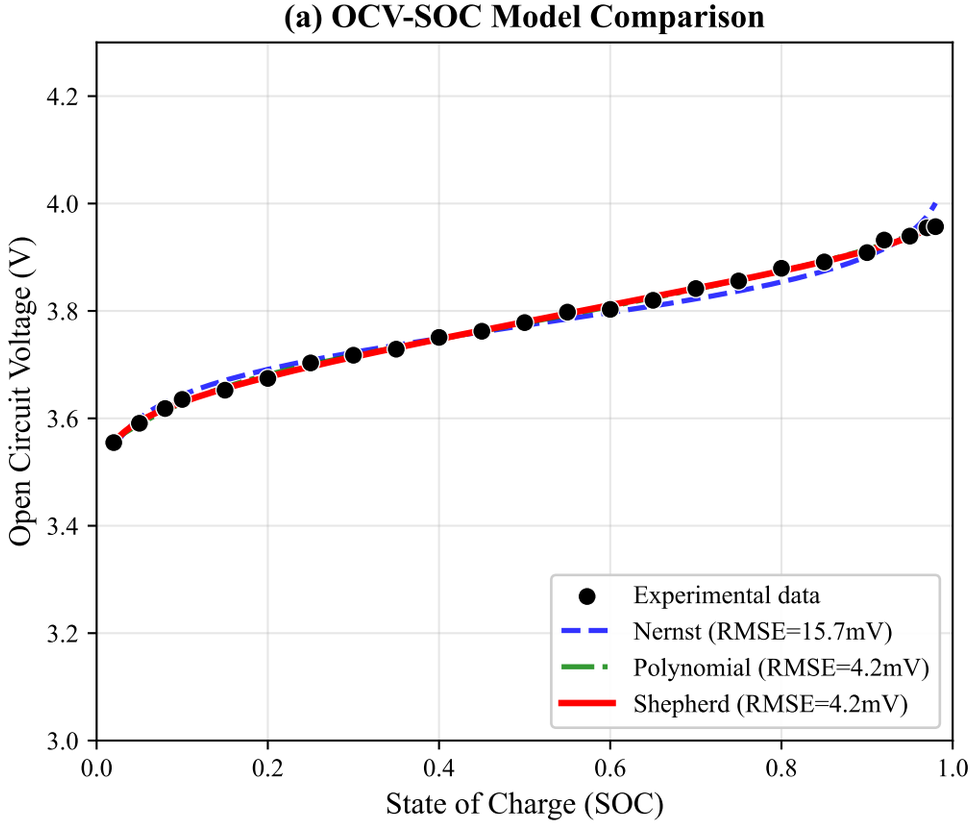
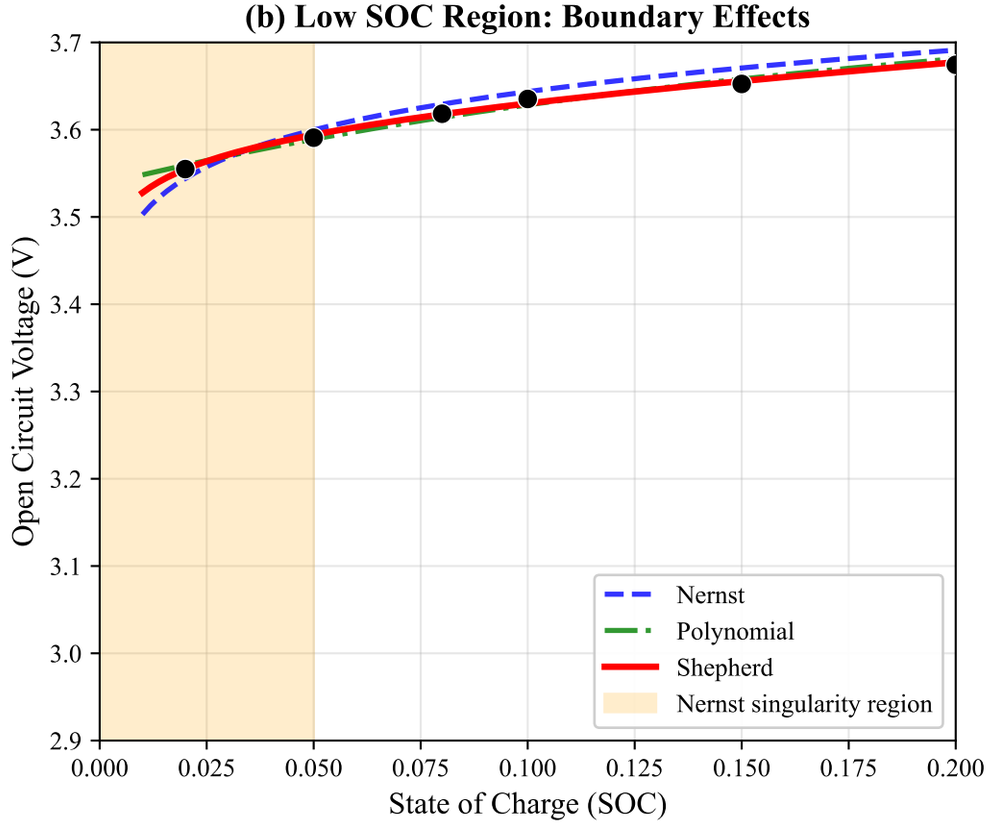
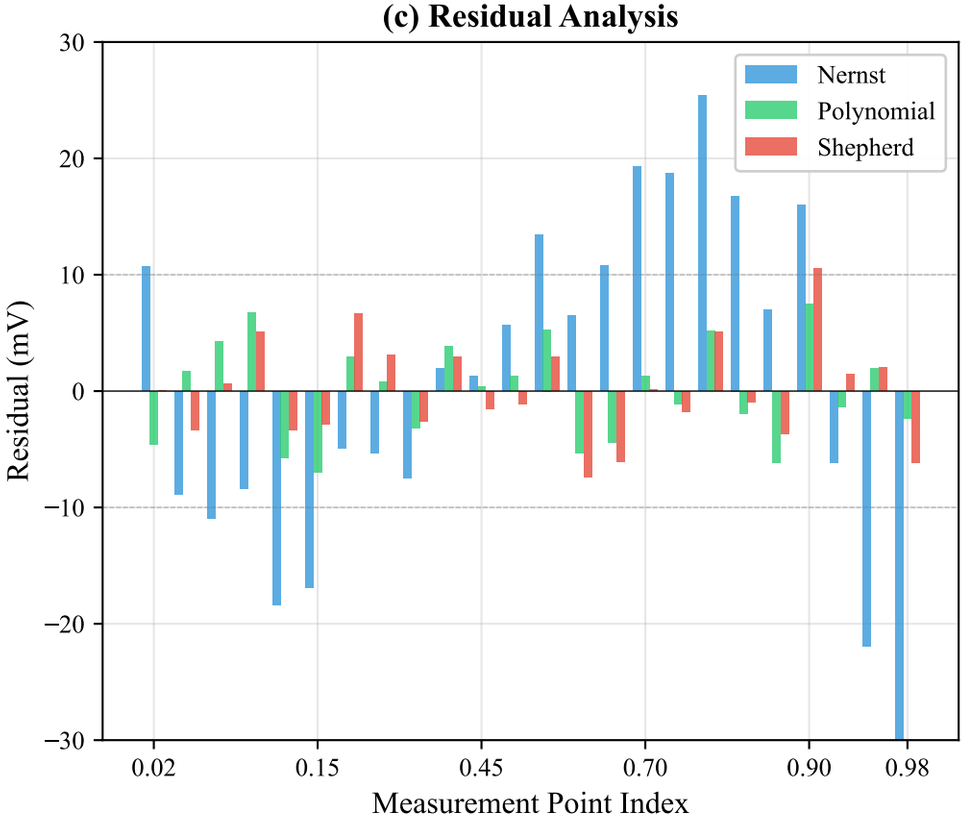
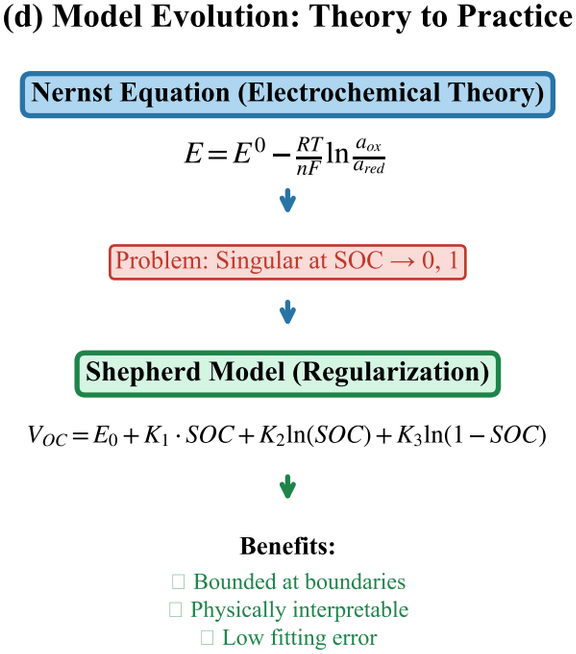

(a) OCV-SOC Model Comparison
Open Circuit Voltage (V)
State of Charge (SOC)
Experimental data
Nernst (RMSE=15.7mV)
Polynomial (RMSE=4.2mV)
Shepherd (RMSE=4.2mV)


(b) Low SOC Region: Boundary Effects
Open Circuit Voltage (V)
State of Charge (SOC)
Nernst
Polynomial
Shepherd
Nernst singularity region


(c) Residual Analysis
Residual (mV)
Measurement Point Index
Nernst
Polynomial
Shepherd


(d) Model Evolution: Theory to Practice
Nernst Equation (Electrochemical Theory)
$E = E^0 - \frac{RT}{nF} \ln \frac{a_{ox}}{a_{red}}$
Problem: Singular at SOC → 0, 1
Shepherd Model (Regularization)
$V_{OC} = E_0 + K_1 \cdot SOC + K_2 \ln(SOC) + K_3 \ln(1 - SOC)$
Benefits:
Bounded at boundaries
Physically interpretable
Low fitting error